The Applicability of Traditional Quantitative Risk Models in Financial Markets: Are Alternative Approaches More Effective for Cryptocurrencies?

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Declaration

I, **Aman Patel**, declare that this dissertation is my own work, conducted under the supervision of **Babar Syed**. All sources used or referred to in this dissertation have been properly acknowledged and cited. Any assistance received in preparing this dissertation has been acknowledged in the acknowledgments section.

I understand that any form of plagiarism, including the unacknowledged use of another author's work, is a serious academic offence and may result in disciplinary action by the university.

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Abstract

This dissertation evaluates the effectiveness of traditional quantitative risk models, Value-at-Risk (VaR) and GARCH, within cryptocurrency markets.

Through empirical comparison with alternative approaches, including machine learning and sentiment analysis, the study finds that conventional models are inadequate for capturing the volatility and behavioural complexity of digital assets. Deep learning models, especially LSTM networks enriched with sentiment data, deliver superior predictive accuracy and adaptability.

The findings mark the necessity of integrating advanced, data-driven frameworks in risk management practices for contemporary, sentiment-sensitive financial environments.

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1 Introduction

1.1 Overview of Financial Risk Management

Financial markets involve exposure to diverse risks, including market, credit, liquidity, and operational risks. Effective risk management is essential to maintain financial stability, guide investment decisions, and comply with regulatory standards. In the past, financial institutions used quantitative frameworks such as Value-at-Risk (VaR) and Generalised Autoregressive Conditional Heteroscedasticity (GARCH) models to quantify and control risk exposure (Engle, 2001). These models acquired widespread acceptance because of their methodological simplicity and adaptability in stable and predictable markets.

However, in recent years, the increased occurrence of market shocks, fundamental crises, and technological advancements has challenged the effectiveness of these traditional models. Their constitutional reliance on historical market data and assumptions of normal distribution have proven inadequate during turbulent market conditions, especially amid fast structural shifts (Ormaniec et al., 2022). Questions emerge regarding their suitability in aggressive financial environments, such as cryptocurrency markets.

1.2 Importance of Risk Models in Stock and Crypto Markets

Quantitative risk models play a fundamental role in maintaining stability and creating confidence in both traditional stock markets and emerging cryptocurrency markets. Institutional investment decisions in traditional stock markets are guided by strong risk models, which inform regulatory oversight and support market efficiency through risk and return assessments. Their effectiveness contributes to investor trust, shaping financial market dynamics and reinforcing macroeconomic stability (Engle, 2001).

Cryptocurrency markets have introduced added complexity. Characterised by decentralisation, continuous trading cycles and pronounced volatility, cryptocurrencies differ from traditional financial assets (Babiak and Bianchi, 2023). This distinct nature challenges existing

risk management strategies and requires innovative frameworks capable of capturing fast, non-linear price movements and unique market characteristics, such as fragmented liquidity and sentiment-driven volatility (Zekiye et al., 2023). Understanding the effectiveness of traditional quantitative models within crypto markets is needed to enhance market resilience and informing regulatory frameworks.

1.3 Why Traditional Models May Be Inadequate for Cryptocurrencies

Despite the proven success of traditional quantitative risk models in established financial markets, their effectiveness in cryptocurrency markets remains uncertain. As noted by Soloviev et al. (2019), "cryptocurrency prices exhibit such complex volatility characteristics as nonlinearity and uncertainty, which are difficult to forecast and any results obtained are uncertain. Therefore, cryptocurrency price prediction remains a huge challenge" (p. 88). These characteristics create conditions divergent from the stable environments assumed by traditional models, which rely on historical data and normal distribution assumptions (So and Yu, 2006).

Models such as VaR and GARCH underestimate risks posed by extreme price movements and sudden liquidity disruptions common in digital assets (Xia, 2023). Traditional methodologies cannot account for structural market features unique to cryptocurrencies, such as fragmented trading environments, continuous trading cycles, and the decentralised nature of transactions (Zekiye et al., 2023). Hence, the research gap on applying traditional models which requires further exploration of alternative or hybrid approaches adapted for managing cryptocurrency risks. As Mashrur et al. (2020) note, "as the task relies heavily on information-driven decision making, machine learning is a promising source for new methods and technologies. In recent years, we have seen increasing adoption of machine learning methods for various risk management tasks" (p. 203203)

1.4 Research Objectives

This dissertation aims to evaluate the effectiveness of traditional quantitative risk models in cryptocurrency markets and to explore whether alternative approaches offer superior accuracy and adaptability. The research will address the following objectives:

- Evaluate the limitations of traditional quantitative risk models (VaR and GARCH) when applied to cryptocurrency markets.
- Identify the features of cryptocurrencies that challenge the foundational assumptions behind traditional risk models.
- Assess the comparative predictive accuracy and reliability of alternative methods (machine learning and sentiment analysis) relative to traditional risk models.
- Explore the viability of hybrid risk management frameworks that combine elements of traditional and alternative models, providing a more comprehensive solution for cryptocurrency markets.

1.5 Structure of the Dissertation

The dissertation begins by reviewing relevant literature to examine traditional quantitative risk management models. Follows on with the discussion of emerging alternative methods, such as machine learning and sentiment analysis, that may address these identified gaps. The research outlines a methodology used to evaluate the predictive effectiveness of traditional and alternative models. Empirical results and analyses follow, providing a comprehensive assessment of their respective strengths and weaknesses. The conclusion summarises key findings, discussing their implications for financial risk management practices, and suggesting potential avenues for future research.

2 Methodology

2.1 Approach

The analysis draws on a comparative research design that evaluates the performance of established risk models, such as Value-at-Risk (VaR) and GARCH, against selected machine learning techniques and sentiment-based models. The study's aim is to determine whether traditional models remain viable in high-volatility digital assets, or whether more adaptive, data-driven approaches offer improved predictive accuracy and responsiveness.

To achieve this, the chapter details the research design, the data sources employed, and the procedures followed in processing, modelling, and evaluating the data. It also considers relevant limitations and ethical considerations.

2.2 Research Design

The design is structured around two quantitative key aspects.

- Evaluate traditional models using historical time series data from Bitcoin.
- Apply machine learning techniques and sentiment analysis to the dataset, allowing direct comparison across predictive performance metrics.

Model performance is evaluated using established statistical measures such as Mean Squared Error (MSE), Value-at-Risk exceedance frequency, and out-of-sample forecasting accuracy. These indicators allow us to quantify each model's predictive reliability and responsiveness to volatility spikes. Models are tested under different market periods across structures to check for accuracy.

The quantitative design allows for replicable, data-driven conclusions grounded in statistical evaluation rather than interpretive judgement. This study supports a transparent assessment of

model applicability in growing financial market conditions, focusing on cryptocurrencies as the primary high-volatility case study.

2.3 Data Collection and Sources

Given the volatility and complexity present in cryptocurrency markets, careful selection and validation of data sources is important. Three primary datasets are employed in this research: cryptocurrency market data, traditional stock market benchmark data, and sentiment-based indicators.

Cryptocurrency market data comprises historical price and trading volume records pulled from reputable exchanges via platforms such as CoinGecko and CryptoCompare. The study focuses on major cryptocurrencies, including Bitcoin (BTC), Ethereum (ETH). The period selected spans from January 2020 to December 2024, focusing on the post-COVID era characterised by extreme volatility, structural market changes, and emerging new trading behaviours. This timeframe aligns with the core aim of assessing traditional risk models under stressed and growing market conditions.

To perform comparative benchmarking, stock market data (historical daily closing prices and volatility figures) from traditional indices, such as the S&P 500 and FTSE 100, has been sourced from Investing.com and Yahoo Finance historical datasets. These indices are benchmarks, allowing an assessment of model performance in lower volatility markets.

Sentiment data, critical for assessing the predictive capabilities of sentiment-based models, is sourced from specialised databases such as Crypto Fear & Greed Index, Santiment, and LunarCrush. The research employs sentiment indicators derived from social media platforms, including Twitter and Reddit, capturing real-time investor sentiment and news-driven market reactions. These datasets provide valuable insights into behavioural influences on cryptocurrency price movements, enhancing the accuracy and responsiveness of alternative risk models.

In this study the Crypto Fear & Greed Index, obtained via the publicly available API from Alternative.me is used. This index quantifies market sentiment on a scale from 0 (Extreme Fear) to 100 (Extreme Greed) and aggregates multiple indicators such as volatility, volume, and

social media trends. This data supports sentiment-driven modelling by providing a consistent and interpretable time series of investor behaviour.

To ensure the consistency of the data used, all datasets underwent a pre-processing stage. This included checks for data completeness, removal of duplicate entries, and treatment of missing values using appropriate imputation techniques. Outliers were identified and adjusted where necessary to avoid misinterpretation in model estimation. Where possible, data points were cross-verified across multiple sources to validate accuracy. These steps were taken to make sure subsequent modelling and analysis would be based on clean, high-quality inputs.

2.4 Analytical Methods and Models

Models selected reflect established methodologies recognised in financial risk management, compared with emerging machine learning and sentiment-based technique with shown potential in predicting complex market behaviours are chosen.

The first category of models comprises traditional risk assessment frameworks. Research evaluates Value-at-Risk (VaR) and Generalised Autoregressive Conditional Heteroscedasticity (GARCH). The VaR model is implemented using both historical simulation and parametric approaches, allowing comparative assessments of its predictive capacity under varied distributional assumptions. The Generalised Autoregressive Conditional Heteroscedasticity (GARCH) model is employed to measure and forecast volatility, capturing conditional variance and volatility clustering effects that characterise cryptocurrency markets. The theoretical foundation for these models originates from Engle's (1982) introduction of the ARCH framework, which allows the conditional variance of a time series to depend on past realisations, thus capturing the phenomenon of volatility clustering.

The second category involves alternative approaches, machine learning algorithms, and sentiment based predictive models. Machine learning models include neural networks (Long Short-Term Memory (LSTM) recurrent neural networks because of their effectiveness in modelling sequential data), support vector machines (SVMs), and random forests. These methods are used because of their ability to handle non-linear relationships, high-dimensional datasets, and complex interaction terms common in cryptocurrency data. The machine learning models follow

a structured method involving dataset splitting into training, validation, and test sets. Hyperparameters are tuned using grid search and cross-validation techniques to optimise predictive performance while mitigating overfitting.

Sentiment analysis is incorporated through natural language processing (NLP) techniques applied to textual data from social media and news sources. Indicators derived from NLP analysis are incorporated into predictive models to assess their added explanatory power concerning market volatility and returns. Measures such as sentiment polarity scores and the Crypto Fear & Greed Index are quantified, integrated as additional explanatory variables within predictive models to enhance their responsiveness to investor sentiment and news events.

All models undergo validation using multiple criteria to compare predictive performance. Forecasting accuracy is assessed using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). The comparative evaluation framework ensures that model performance is assessed with good validity, promoting clear conclusions regarding their suitability for managing cryptocurrency market risks.

2.5 Limitations and Ethical Considerations

While the methodological framework adopted ensures clarity, it is important to acknowledge certain limitations. The study's reliance on historical data introduces constraints, given the changing nature of cryptocurrency markets. Past performance and historical volatility patterns may not capture future market dynamics or sudden regime shifts, limiting predictive accuracy.

Sentiment analysis methods, though powerful, can be subject to biases and inaccuracies present in social media and news-based sentiment scoring. The subjective nature of language, presence of noise, and potential manipulation or amplification of sentiment data pose further limitations that could influence model outcomes.

The Fear & Greed Index is based on composite indicators, which may introduce generalisation bias. However, its public accessibility and established use in financial commentary justify its application in academic research. All sentiment sources used were public and free from ethical concerns regarding personal data.

From an ethical standpoint, the research maintains transparency in data handling and model selection. All datasets are available publicly or licensed, and no sensitive or private user information is utilised. Ethical considerations were addressed by ensuring responsible data management practices, safeguarding transparency, integrity, and reproducibility throughout the analysis.

3 Analysis and Findings

3.1 Overview

The analysis draws on key statistical metrics such as Value-at-Risk (VaR) exceedance frequency, volatility forecasts, and prediction error rates including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).

Models are evaluated across different timeframes and market conditions, allowing for an assessment of their predictive reliability and robustness. In addition, the influence of sentiment-based indicators on forecast accuracy is explored to determine the added value of integrating behavioural data into risk estimation.

Visualisations and summary tables are used throughout to illustrate differences in model performance. These findings are then interpreted to draw insights into applying each modelling approach for managing risk in high-volatility, decentralised financial environments.

3.2 Results from Traditional Models

The analysis of traditional quantitative risk models, Value-at-Risk (VaR) and Generalised Autoregressive Conditional Heteroscedasticity (GARCH), provides insights into their applicability to cryptocurrency markets compared to traditional financial indices. Table 1 summarises the VaR results at the 95% and 99% confidence levels for Bitcoin, Ethereum, the Dow Jones Index, and the S&P 500.

Table 3.1: VaR Exceedance Summary

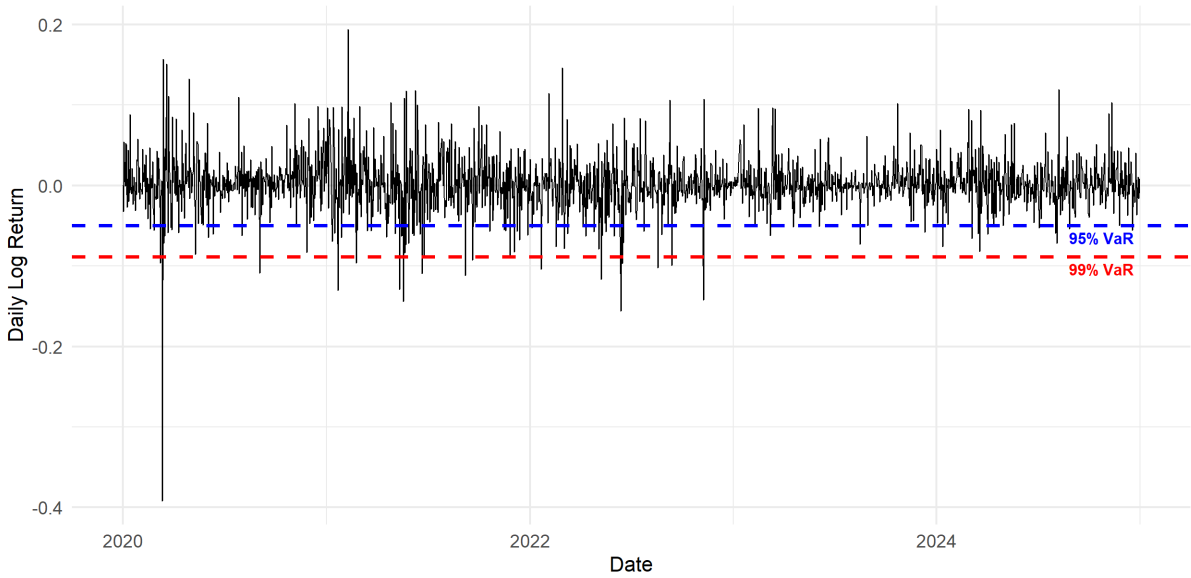
Asset	**VaR 95%**	**Exceedance (%)**	**VaR 99%**	**Exceedance (%)**
Bitcoin	-0.04997	5.04	-0.08853	1.04
Ethereum	-0.06388	5.04	-0.11196	1.04
Dow Jones	-0.01712	5.00	-0.03562	1.03
S&P 500	-0.01866	5.01	-0.03592	1.03

Source: Investing.com

The VaR backtesting shows consistent exceedance rates close to the theoretical thresholds for traditional indices (Dow Jones and S&P 500), demonstrating reliable performance in established markets.

In contrast, Bitcoin and Ethereum show larger VaR values because of substantial volatility, showing that cryptocurrency market models require higher risk thresholds. ”Financial uncertainty is the one that is most easily and precisely measured at high frequency and for a variety of levels of aggregation, and because of this, it has most often been used as a proxy for uncertainty more generally” (Dew-Becker & Giglio, 2024, p. 40). Figure 1 illustrates Bitcoin’s daily returns against historical VaR levels at the 95% and 99% confidence intervals, revealing instances where actual losses exceeded the predicted thresholds.

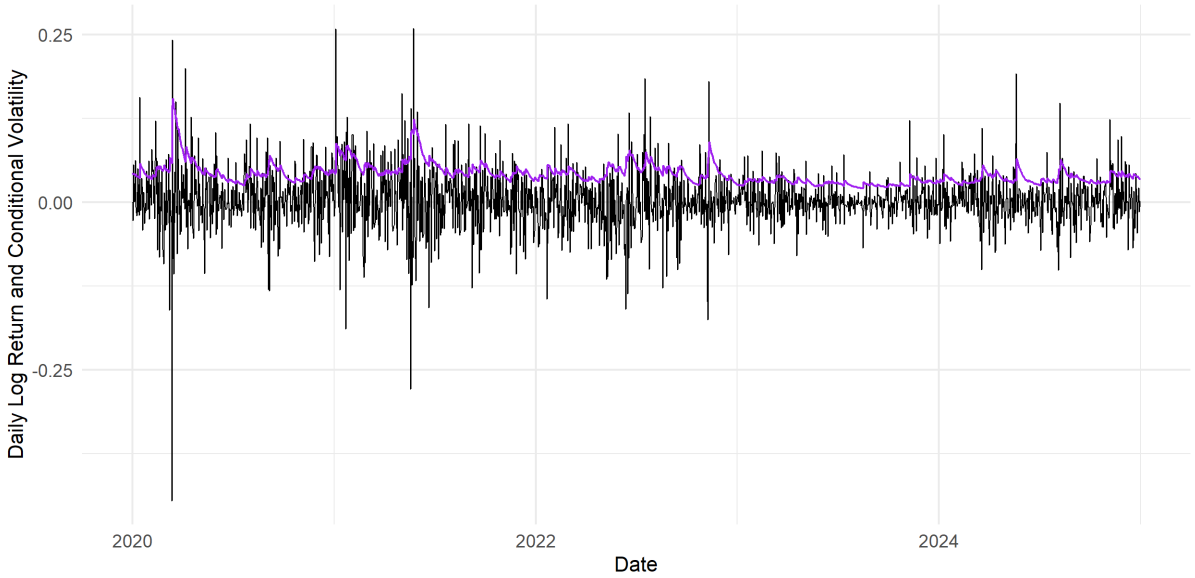
Figure 3.1: Bitcoin Daily Returns vs Historical VaR



Source: Investing.com

The GARCH model further enhances the analysis by addressing volatility clustering common in cryptocurrencies. Ethereum’s daily returns and predicted conditional volatility from the GARCH (1,1) model are presented in Figure 2. The figure illustrates substantial volatility clustering, noticeable during market shocks between 2021 and 2022. The clustering of volatility observed in both traditional and cryptocurrency markets positively marks models such as GARCH, which extend the ARCH approach originally developed by Engle (1982), specifically to model periods of persistently high or low variance.

Figure 3.2: Ethereum Daily Returns and GARCH (1,1) Volatility



Source: Investing.com

The comparative performance of VaR and GARCH across traditional and cryptocurrency markets marks a clear distinction: while both models provide stable and predictable performance for traditional assets, their predictive accuracy diminishes under conditions of extreme volatility typical of cryptocurrency markets. These findings reinforce the idea of adapting traditional models or considering alternative risk frameworks better suited for digital assets.

3.3 Results from Alternative Models

Alternative models, such as machine learning (ML) and deep learning techniques, provide additional insights into cryptocurrency returns compared to traditional econometric models. The performance metrics, summarised in Table 2, show differences in prediction accuracy between these approaches.

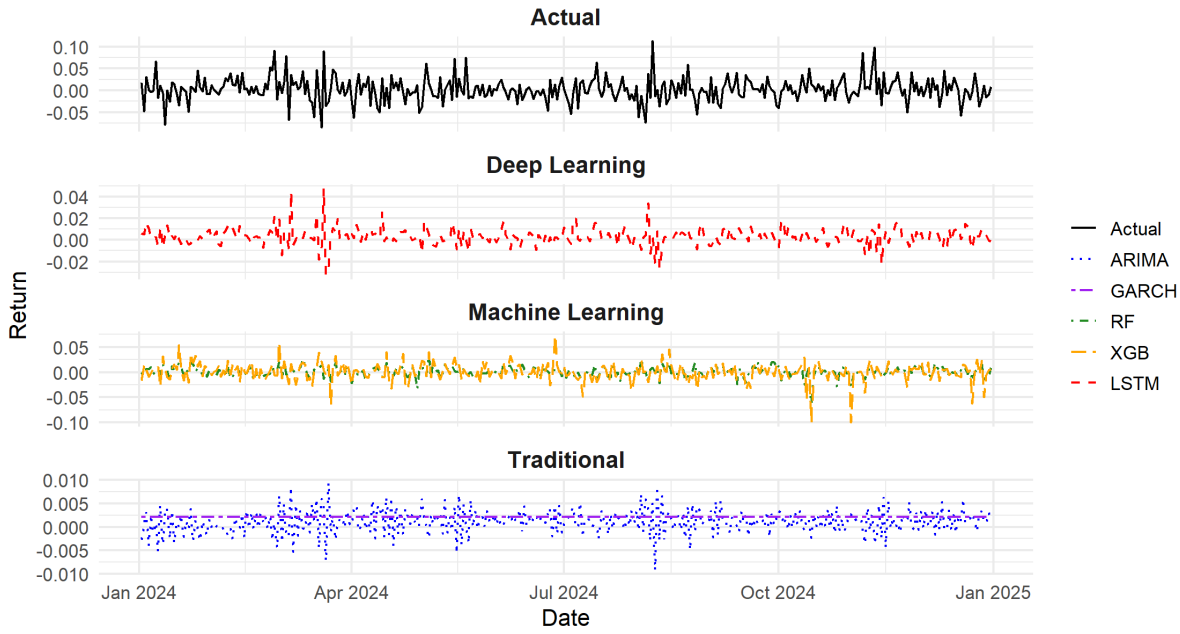
Table 3.2: Model Performance Comparison

Model	MSE	RMSE
ARIMA	0.00076	0.27490
GARCH	0.00076	0.27550
Random Forest	0.00089	0.29770
XGBoost	0.00183	0.04280
LSTM	0.00065	0.02551

Source: Investing.com

The LSTM model shows superior predictive accuracy with the lowest Mean Squared Error (MSE) and Root Mean Squared Error (RMSE), far outperforming traditional ARIMA and GARCH models, and even machine learning approaches, like Random Forest and XGBoost. Figure 3 compares actual Bitcoin returns with predicted returns across traditional, machine learning, and deep learning models, illustrating the accuracy of the LSTM predictions in capturing return volatility and sharp market movements.

Figure 3.3: Predicted vs Actual Bitcoin Returns by Model Group

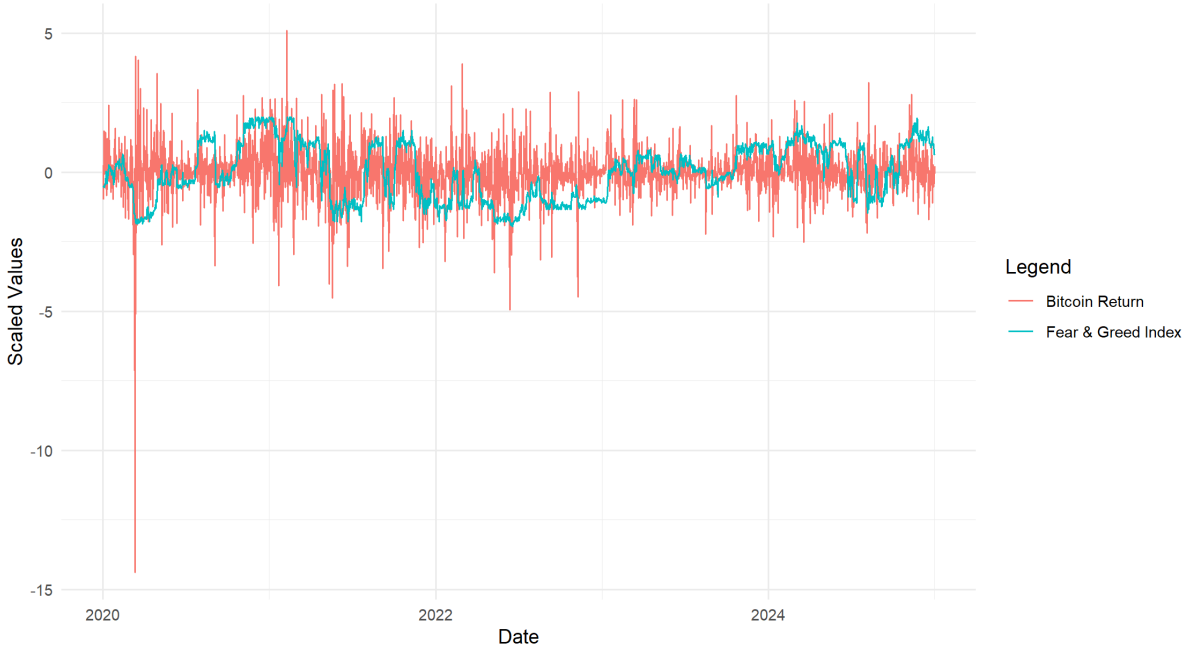


Source: Investing.com

Incorporating sentiment-based data, through the Fear & Greed Index, further enhanced the predictive performance. Figure 4 presents the scaled values of Bitcoin returns and the Fear & Greed Index, demonstrating that sentiment indicators track return volatility, especially during

market downturns and recovery periods. The strong correlation observed supports including sentiment as a meaningful explanatory variable within predictive models.

Figure 3.4: Bitcoin Returns vs Fear & Greed Index



Sources: Investing.com & Alternative.me

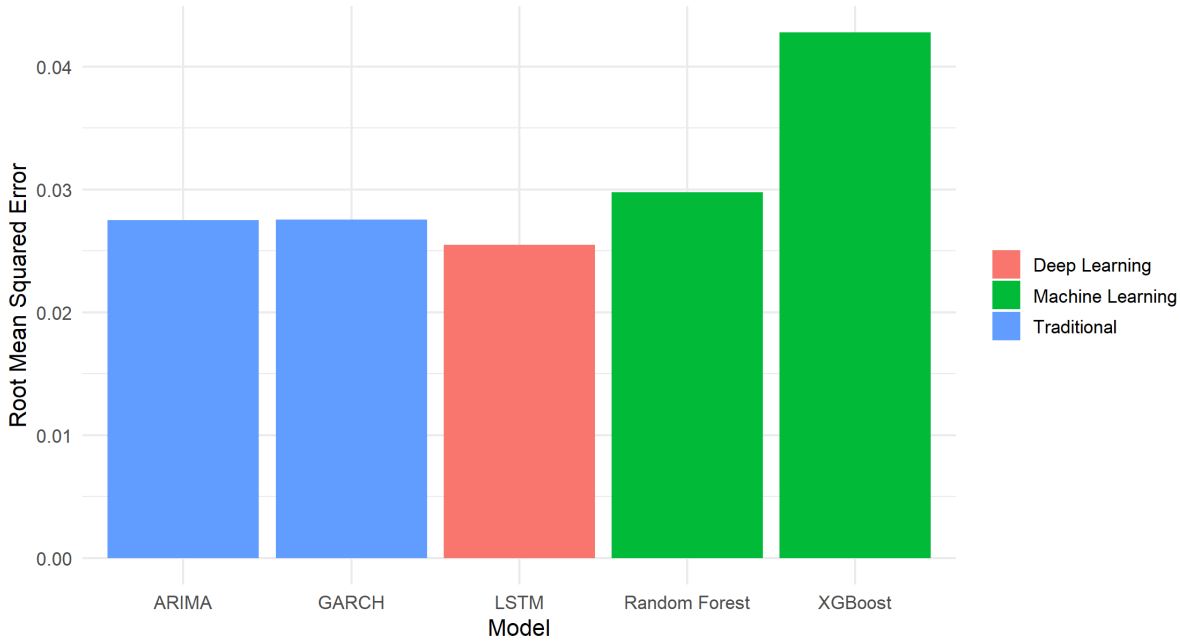
However, during the rolling GARCH volatility estimation, the model failed to converge in several windows, resulting in gaps in the volatility time series. This phenomenon aligns with the existing financial econometrics literature, occurring when the local return series lack sufficient variance or structural characteristics required for stable model fitting. Such convergence issues underline the intrinsic challenges traditional econometric models face in volatile markets, such as cryptocurrencies.

These findings suggest alternative models, such as deep learning, approaches enriched with sentiment data, provide a meaningful improvement in capturing and predicting the often erratic behaviour of cryptocurrency markets.

3.4 Comparative Evaluation

The comparative evaluation shows the relative strengths of the modelling approaches. Figure 5 reports the Root Mean Squared Error (RMSE) across traditional econometric models (ARIMA, GARCH), machine learning models (Random Forest, XGBoost), and the deep learning model (LSTM).

Figure 3.5: Model RMSE Comparison by Type



Source: Investing.com

The LSTM model shows superior performance across different volatility and sentiment regimes, as displayed in Table 3, exhibited by the lowest RMSE values in high, medium, and low volatility conditions, suggesting adaptability to varying market conditions.

Table 3.3: Best Models by Volatility Regime

Volatility Condition	**Model**	**RMSE**
High	LSTM	0.0318
Mid	LSTM	0.0277
Low	LSTM	0.0194

Source: Investing.com

When assessing predictive performance under different sentiment conditions, the LSTM model

again emerges as the strongest performer. It excels in periods of extreme fear and greed, conditions characterised by higher market uncertainty and fast price movements.

Table 3.4: Best Models by Sentiment Regime

Sentiment Condition	**Model**	**RMSE**
Extreme Fear	LSTM	0.0307
Extreme Greed	LSTM	0.0261
Neutral	LSTM	0.0230

Source: Alternative.me

These findings reinforce the effectiveness of deep learning approaches, in particular when they are enriched with sentiment-based inputs, in managing risk and predicting returns in cryptocurrency markets. Whereas, traditional econometric and standard machine learning models displayed higher prediction errors, revealing their limitations in adapting to volatility and sentiment-driven behaviours ingrained in cryptocurrencies.

4 Discussion

4.1 Interpretation of Key Findings

The empirical analysis reveals a distinct divergence in the effectiveness of traditional and alternative risk models within the context of cryptocurrency markets. Value-at-Risk (VaR) and GARCH models show reliable performance for traditional financial indices, where volatility is subdued and market structures are stable. In these settings, backtesting shows exceedance rates remain close to theoretical thresholds, and GARCH captures volatility clustering.

However, the results also show critical limitations of these traditional models when applied to cryptocurrencies. The greater magnitude of VaR values and more frequent exceedance for assets such as Bitcoin and Ethereum accentuate the higher intrinsic risk and the inadequacy of traditional methods under conditions of extreme volatility. Convergence issues in rolling GARCH estimations point to challenges in fitting econometric models in environments characterised by irregular market regimes and abrupt structural breaks.

Introducing machine learning and deep learning models, especially LSTM networks, leads to striking improvements in predictive accuracy. The LSTM model outperforms both traditional econometric and standard machine learning models across various volatility and sentiment regimes, as evidenced by the lowest RMSE values. The capacity of LSTM to capture nonlinear dependencies, combined with integrating sentiment-based inputs, positions it as a potential tool for forecasting returns and managing risk in the erratic cryptocurrency sector. Sentiment analysis, when weighing in indices like the Fear & Greed Index, further enhances the responsiveness of models to abrupt market shifts driven by collective investor behaviour.

These findings imply that alternative, data-driven approaches are better suited to the sentiment-sensitive nature of cryptocurrencies. On the other hand, traditional models provide a useful benchmark and keep their relevance in conventional markets, their predictive limitations are exposed to high-volatility, decentralised environments. For practitioners and regulators, this suggests a need to adopt or at least supplement existing risk management frameworks with machine learning and sentiment analysis techniques to mitigate risk with accuracy in digital

asset markets.

4.2 Implications for Risk Management Practice

The findings of this study have important implications for contemporary risk management in both cryptocurrency and broader financial markets. The limitations observed in traditional models such as VaR and GARCH suggest that sole reliance on these approaches may expose practitioners and institutions to underestimated risks during periods of heightened volatility or market disruption. In cryptocurrencies, where decentralisation, fast trading cycles, and sentiment-driven movements are prevalent, these models are challenged, often failing to expect sharp price swings or structural breaks.

The shown success of machine learning, like the LSTM models augmented with sentiment indicators, points towards the practical value of integrating data-driven techniques into risk assessment frameworks. Such models are more adaptive, capable of processing non-linear patterns and responding to shifts in market sentiment. This adaptability is critical for portfolio managers and risk officers seeking to maintain accurate risk controls in unpredictable environments.

Supervisory bodies and compliance teams should consider the limitations of established quantitative methods when evaluating the resilience of institutions engaged in digital asset trading. The evidence supports a shift towards hybrid risk management practices, where traditional models are supplemented with advanced analytics and real-time sentiment monitoring to better capture the unique risks posed by cryptocurrencies.

4.3 Theoretical and Methodological Reflections

This study contributes to the theoretical discourse by challenging the adequacy of conventional risk modelling frameworks in cryptocurrencies. Results affirm the concerns raised in the literature regarding the limitations of VaR and GARCH models with their reliance on distributional assumptions and historical data, which are violated in decentralised digital asset markets. The

original ARCH process, as introduced by Engle (1982), assumes that the conditional variance evolves as a function of past squared residuals, providing a mechanism for modelling changing volatility over time in financial time series. Empirical evidence aligns with recent calls for a shift towards more flexible and responsive modelling approaches. This approach is supported by Babiak & Bianchi (2023), who find that "a combination of trading frictions and network growth might be the key determinant of risk premiums within the cryptocurrency space" (p. 5), whilst the findings of Soloviev et al. (2019), also show that complexity measures based on Random Matrix Theory can act as effective early warning signals for impending crashes in cryptocurrency markets (pp. 87–88).

The comparative analysis adopted in this dissertation represents a considerable strength, enabling an objective assessment of predictive performance across model types and market conditions. With the integration, sentiment analysis and machine learning shows an emerging consensus in financial econometrics, reiterating the need to account for behavioural and non-linear influences. However, the study recognises its limitations, including the challenges of model interpretability, the risk of overfitting in deep learning approaches, and potential biases ingrained in sentiment data.

4.4 Limitations and Future Research

Several limitations intrinsic to this study should be acknowledged. First, the reliance on historical data may restrict the generalisability of the findings, as past market behaviour does not always predict future trends. Sentiment data, while valuable, is subject to biases and noise, including manipulation or coordinated campaigns on social media, which can affect model reliability. The convergence issues observed with GARCH show methodological challenges when volatility patterns are irregular.

Future research should explore integrating more granular or alternative sentiment measures, such as blockchain network activity or real-time transaction flows, to enhance predictive accuracy. A hybrid or ensemble models that combine the strengths of econometric, machine learning, and behavioural approaches may also offer promise. As demonstrated by Babiak & Bianchi (2023), the IPCA framework can deliver an out-of-sample R^2 of 52% for portfolio re-

turns, outperforming observable factor models (19%) (Table 6). Extending the analysis to a wider range of digital assets and market regimes could further strengthen the evidence base for risk model selection.

As Dew-Becker & Giglio (2024) indicate , understanding financial risk in modern markets requires models capable of capturing nonlinear processes and time-varying higher moments, such as skewness, which are often neglected in standard linear approaches (p. 55).

4.5 Closure

In summary, this study shows that traditional risk models alone do not manage the complexities of cryptocurrency markets. The evidence supports the adoption of adaptive, data-driven approaches, particularly those using machine learning and sentiment analysis. By advancing a more refined framework for risk assessment, this research contributes to both academic debate and practical financial risk management.

5 Conclusion

5.1 Summary of Key Findings

This dissertation provides a comparative assessment of traditional and alternative quantitative risk models in cryptocurrency markets. The empirical analysis shows that while Value-at-Risk (VaR) and GARCH models perform for conventional financial indices, they cannot capture the extreme volatility, heavy tails, and irregular dynamics characteristic of digital assets. This limitation is acute in periods marked by price swings and market contagion in cryptocurrency trading.

The findings are consistent with recent literature indicating the inadequacy of traditional risk frameworks for cryptocurrencies. As Yermack (2013) recognises, Bitcoin's volatility, lack of correlation with other major asset classes, and speculative price movements "make bitcoin useless for risk management and exceedingly difficult for its owners to hedge" (Yermack, 2013, p. 2). This shows why conventional approaches, reliant on assumptions of normality and stable market structure, often fail in crypto markets.

The work of Fry and Cheah (2016) provides empirical evidence of market phenomena absent from traditional finance. They report "during the negative bubble there is a spillover from Ripple which then exacerbates the subsequent falls in Bitcoin" (Fry & Cheah, 2016, p. 349). Such dynamics further stress the limitations of established risk models when applied to cryptocurrencies.

In contrast, this study finds that machine learning and sentiment-enriched models, specifically LSTM networks, deliver superior predictive accuracy and adaptability. These alternative approaches outperform traditional models during periods of market turbulence and sentiment-driven volatility, suggesting their important value for risk management in digital assets.

5.2 Contributions, Recommendations and Future Research

This research advances the literature by providing empirical evidence that traditional risk models, such as VaR and GARCH, are often inadequate in cryptocurrencies. The study exhibits the unique features of digital asset markets, such as high volatility, lack of correlation with other asset classes, and the prevalence of both speculative and negative bubbles, require a more adaptive approach to risk management.

Based on these findings, the following recommendations are proposed:

- For practitioners and regulators: Move beyond sole reliance on conventional risk models. Integrate machine learning techniques and real-time sentiment analysis to improve risk detection and response in digital asset markets.
- For portfolio managers: Employ hybrid risk management strategies which can identify not just routine volatility but also sentimental speculation and negative bubble formation.
- For policy development: Update risk assessment frameworks to reflect emerging empirical research and advances in modelling techniques.

For future research, it is recommended to:

- Investigate the performance of alternative models across additional cryptocurrencies and broader market conditions.
- Develop modelling approaches which can capture negative bubbles and contagion dynamics, as outlined by Fry & Cheah (2016).
- Explore richer sentiment and on-chain datasets with ensemble methods combining econometric and machine learning approaches, to further enhance risk forecasting and management.

5.3 Final Remarks

The findings of this dissertation signify the urgent need for risk models that can adapt to the complexities and unique behaviours of cryptocurrency markets. As both the present analysis and leading studies make clear, traditional models alone are no longer sufficient for environments defined by extreme volatility, speculative bubbles, and cross-asset contagion.

Looking ahead, the effective management of financial risk in digital markets will depend on the willingness of practitioners and researchers to adopt innovative frameworks, which incorporate machine learning, sentiment analysis, and complexity science. Only through such adaptation can the field keep pace with the growing risks and opportunities of the cryptocurrency era.

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A Appendix

A.1 Raw Data

Bitcoin Historical Results Price Data

The full historical Bitcoin price data used in the analysis is provided as a supplementary file. For reference, the first 10 rows of the dataset are displayed in Table A.1.

Date	Price	Open	High	Low	Vol.	Change %
31/12/2024	93,557.2	92,777.2	96,163.4	92,036.2	74.85K	0.84%
30/12/2024	92,779.8	93,718.7	94,936.4	91,522.3	112.43K	-1.00%
29/12/2024	93,716.3	95,282.6	95,315.4	93,026.7	47.71K	-1.65%
28/12/2024	95,284.5	94,274.9	95,684.3	94,124.7	32.97K	1.07%
27/12/2024	94,275.9	95,776.4	97,243.3	93,472.8	85.12K	-1.57%
26/12/2024	95,777.7	99,389.4	99,922.5	95,193.3	74.60K	-3.64%
25/12/2024	99,391.3	98,661.9	99,514.1	97,651.9	39.80K	0.73%
24/12/2024	98,668.1	94,852.8	99,446.6	93,566.2	69.86K	4.02%
23/12/2024	94,852.1	95,184.8	96,517.8	92,543.2	133.24K	-0.35%
22/12/2024	95,183.8	97,281.6	97,422.2	94,265.2	100.64K	-2.13%

Table A.1: Sample of the Bitcoin historical price data used in this dissertation.

Ethereum Historical Results Price Data

The full historical Ethereum price data used in the analysis is provided as a supplementary file. For reference, the first 10 rows of the dataset are displayed in Table A.2.

Dow Jones Industrial Average Historical Results Price Data

The full historical Dow Jones Industrial Average price data used in the analysis is provided as a supplementary file. For reference, the first 10 rows of the dataset are displayed in Table A.3.

Date	Price	Open	High	Low	Vol.	Change %
31/12/2024	3,337.80	3,361.99	3,449.27	3,317.39	333.44K	-0.72%
30/12/2024	3,361.99	3,355.97	3,436.83	3,306.11	494.99K	0.18%
29/12/2024	3,355.97	3,404.53	3,412.32	3,327.45	176.43K	-1.43%
28/12/2024	3,404.59	3,334.23	3,427.65	3,322.42	196.38K	2.11%
27/12/2024	3,334.19	3,335.96	3,438.85	3,309.03	378.00K	-0.05%
26/12/2024	3,335.96	3,496.50	3,512.77	3,305.53	360.40K	-4.59%
25/12/2024	3,496.49	3,493.33	3,546.29	3,442.61	259.25K	0.09%
24/12/2024	3,493.31	3,422.01	3,539.54	3,359.77	352.36K	2.08%
23/12/2024	3,422.01	3,281.58	3,463.58	3,220.10	547.74K	4.28%
22/12/2024	3,281.58	3,338.72	3,401.96	3,223.39	400.57K	-1.71%

Table A.2: Sample of the Ethereum historical price data used in this dissertation.

Date	Price	Open	High	Low	Vol.	Change %
23/05/2025	41,603.07	41,525.70	41,788.61	41,354.09	497.07M	-0.61%
22/05/2025	41,859.09	41,763.68	42,090.42	41,714.43	476.08M	0.00%
21/05/2025	41,860.44	42,354.46	42,439.27	41,785.22	607.90M	-1.91%
20/05/2025	42,677.24	42,735.11	42,800.04	42,485.37	441.54M	-0.27%
19/05/2025	42,792.07	42,542.81	42,842.04	42,337.71	536.84M	0.32%
16/05/2025	42,654.74	42,356.20	42,663.68	42,232.56	621.26M	0.78%
15/05/2025	42,322.75	41,777.98	42,351.42	41,777.98	736.43M	0.65%
14/05/2025	42,051.06	42,150.09	42,254.75	41,952.86	660.23M	-0.21%
13/05/2025	42,140.43	42,507.33	42,507.33	42,132.04	729.87M	-0.64%
12/05/2025	42,410.10	41,899.05	42,418.05	41,899.05	653.82M	2.81%

Table A.3: Sample of the Dow Jones Industrial Average historical price data used in this dissertation.

S&P 500 Historical Results Price Data

The full historical S&P 500 price data used in the analysis is provided as a supplementary file. For reference, the first 10 rows of the dataset are displayed in Table A.4.

A.2 Code Snippets

R Code: Comparison of Historical Volatility Cryptocurrencies vs. Stock Indices

Date	Price	Open	High	Low	Vol.	Change %
31/12/2024	5,881.63	5,919.74	5,929.74	5,868.86	—	-0.43%
30/12/2024	5,906.94	5,920.67	5,940.79	5,869.16	—	-1.07%
27/12/2024	5,970.84	6,006.17	6,006.17	5,932.95	—	-1.11%
26/12/2024	6,037.59	6,024.97	6,049.75	6,007.37	—	-0.04%
24/12/2024	6,040.04	5,984.63	6,040.10	5,981.44	—	1.10%
23/12/2024	5,974.07	5,940.25	5,978.25	5,902.57	—	0.73%
20/12/2024	5,930.85	5,842.00	5,982.06	5,832.30	—	1.09%
19/12/2024	5,867.08	5,912.71	5,935.52	5,866.07	—	-0.09%
18/12/2024	5,872.16	6,047.65	6,070.67	5,867.79	—	-2.95%
17/12/2024	6,050.61	6,052.55	6,057.68	6,035.19	—	-0.39%

Table A.4: Sample of the S&P 500 historical price data used in this dissertation.

```r
# Load required packages
library(dplyr)
library(lubridate)
library(zoo)
library(tidyr)
library(ggplot2)

# Define a function to clean each dataset
clean_data <- function(df, date_format = "%m/%d/%Y") {
  df <- df %>%
    # Select only the needed columns (Date and Price)
    select(Date, Price) %>%
    # Remove commas from Price and convert to numeric
    mutate(
      Price = as.numeric(gsub(",", "", Price)),
      Date = as.Date(Date, format = date_format)
    ) %>%
    arrange(Date)  # Ensure the data is in ascending order
  return(df)
}

# Read in your CSV files
btc <- read.csv("Bitcoin 2020-24.csv", stringsAsFactors = FALSE)
```

```r
eth <- read.csv("Ethereum 2020-24.csv", stringsAsFactors = FALSE)
sp500 <- read.csv("S&P 500 2020-24.csv", stringsAsFactors = FALSE)
dow <- read.csv("Dow Jones 2020-24.csv", stringsAsFactors = FALSE)

# Clean each dataset
btc_clean   <- clean_data(btc, date_format = "%m/%d/%Y")
eth_clean   <- clean_data(eth, date_format = "%m/%d/%Y")
sp500_clean <- clean_data(sp500, date_format = "%d/%m/%Y")
dow_clean   <- clean_data(dow, date_format = "%m/%d/%Y")

# Function to compute daily log returns
compute_returns <- function(df) {
  df %>%
    mutate(Return = c(NA, diff(log(Price)))) %>%
    filter(!is.na(Return))
}

# Compute log returns for each asset
btc_returns   <- compute_returns(btc_clean)
eth_returns   <- compute_returns(eth_clean)
sp500_returns <- compute_returns(sp500_clean)
dow_returns   <- compute_returns(dow_clean)

# Function to compute rolling 30-day annualised volatility
compute_volatility <- function(returns_series, window = 30) {
  # Calculate rolling standard deviation and annualise it (using 252 trading days)
  roll_sd <- rollapply(returns_series, width = window, FUN = sd, fill = NA, align = "right")
  volatility <- roll_sd * sqrt(252)
  return(volatility)
}
```

```r

# Apply the rolling volatility function to each asset
btc_returns$Volatility   <- compute_volatility(btc_returns$Return, window = 30)
eth_returns$Volatility   <- compute_volatility(eth_returns$Return, window = 30)
sp500_returns$Volatility <- compute_volatility(sp500_returns$Return, window = 30)
dow_returns$Volatility   <- compute_volatility(dow_returns$Return, window = 30)

# Prepare data for plotting by selecting Date and Volatility, and adding an Asset identifier
btc_vol   <- btc_returns %>% select(Date, Volatility) %>% mutate(Asset = "Bitcoin")
eth_vol   <- eth_returns %>% select(Date, Volatility) %>% mutate(Asset = "Ethereum")
sp500_vol <- sp500_returns %>% select(Date, Volatility) %>% mutate(Asset = "S&P 500")
dow_vol   <- dow_returns %>% select(Date, Volatility) %>% mutate(Asset = "Dow Jones")

# Combine all data into one data frame
vol_data <- bind_rows(btc_vol, eth_vol, sp500_vol, dow_vol)

# Plot the 30-Day Rolling Annualised Volatility using ggplot2
ggplot(vol_data, aes(x = Date, y = Volatility, colour = Asset)) +
  geom_line(size = 1) +
  labs(title = "30-Day Rolling Annualised Volatility (2020-2024)",
       x = "Date",
       y = "Annualised Volatility",
       colour = "Asset") +
  theme_minimal() +
```

```
  theme(legend.position = "bottom",
        plot.title = element_text(hjust = 0.5))
```

Listing A.1: R script for Comparison of Historical Volatility

R Code: BTC VAR and ETH Vol Predicted GARCH

```
# Load required packages
library(readr)
library(dplyr)
library(xts)
library(PerformanceAnalytics)
library(quantmod)
library(ggplot2)
library(rugarch)

# Load cleaned CSVs
btc <- read_csv("BTC_Clean.csv")
eth <- read_csv("ETH_Clean.csv")
djia <- read_csv("DJIA_Clean.csv")
spx <- read_csv("SPX_Clean.csv")

# Convert to Date and sort
btc <- btc %>% mutate(Date = as.Date(Date)) %>% arrange(Date)
eth <- eth %>% mutate(Date = as.Date(Date)) %>% arrange(Date)
djia <- djia %>% mutate(Date = as.Date(Date)) %>% arrange(Date)
spx <- spx %>% mutate(Date = as.Date(Date)) %>% arrange(Date)

# VAR and Exceedance

# Calculate daily returns
btc$return <- dailyReturn(xts(btc$Close, order.by = btc$Date))
eth$return <- dailyReturn(xts(eth$Close, order.by = eth$Date))
djia$return <- dailyReturn(xts(djia$Close, order.by = djia$Date))
```

```r
spx$return <- dailyReturn(xts(spx$Close, order.by = spx$Date))

# Bitcoin
btc_returns <- na.omit(btc$return)
btc_var_95 <- quantile(btc_returns, probs = 0.05)
btc_var_99 <- quantile(btc_returns, probs = 0.01)
btc_exc_95 <- mean(btc_returns < btc_var_95) * 100
btc_exc_99 <- mean(btc_returns < btc_var_99) * 100

# Ethereum
eth_returns <- na.omit(eth$return)
eth_var_95 <- quantile(eth_returns, probs = 0.05)
eth_var_99 <- quantile(eth_returns, probs = 0.01)
eth_exc_95 <- mean(eth_returns < eth_var_95) * 100
eth_exc_99 <- mean(eth_returns < eth_var_99) * 100

# Dow Jones
djia_returns <- na.omit(djia$return)
djia_var_95 <- quantile(djia_returns, probs = 0.05)
djia_var_99 <- quantile(djia_returns, probs = 0.01)
djia_exc_95 <- mean(djia_returns < djia_var_95) * 100
djia_exc_99 <- mean(djia_returns < djia_var_99) * 100

# S&P 500
spx_returns <- na.omit(spx$return)
spx_var_95 <- quantile(spx_returns, probs = 0.05)
spx_var_99 <- quantile(spx_returns, probs = 0.01)
spx_exc_95 <- mean(spx_returns < spx_var_95) * 100
spx_exc_99 <- mean(spx_returns < spx_var_99) * 100

# Combine into a single table
var_table <- data.frame(
  Asset = c("Bitcoin", "Ethereum", "Dow Jones", "S&P 500"),
```

```
  VaR_95 = round(c(btc_var_95, eth_var_95, djia_var_95, spx_var_95), 5),
  Exceed_95 = round(c(btc_exc_95, eth_exc_95, djia_exc_95, spx_exc_95), 2),
  VaR_99 = round(c(btc_var_99, eth_var_99, djia_var_99, spx_var_99), 5),
  Exceed_99 = round(c(btc_exc_99, eth_exc_99, djia_exc_99, spx_exc_99), 2)
)

print(var_table)

#Figure 1 - Time Series Plot of BTC Returns vs VaR 95 and 99

# Prepare plot data
# Create a new dataframe with regular types
btc_plot <- data.frame(
  Date = index(btc$return),
  return = as.numeric(btc$return)
)

btc_var_95 <- quantile(btc_plot$return, probs = 0.05)
btc_var_99 <- quantile(btc_plot$return, probs = 0.01)


label_x_pos <- max(btc_plot$Date) - 10
# Plot
ggplot(btc_plot, aes(x = Date)) +
  geom_line(aes(y = return), colour = "black", linewidth = 0.3) +
  geom_hline(yintercept = btc_var_95, linetype = "dashed", colour = "blue", linewidth = 1) +
  geom_hline(yintercept = btc_var_99, linetype = "dashed", colour = "red", linewidth = 1) +
  annotate("text", x = label_x_pos, y = btc_var_95 - 0.015,
```

```
            label = "95% VaR", hjust = 1, colour = "blue", size = 3.5, fontface = "bold") +
  annotate("text", x = label_x_pos, y = btc_var_99 - 0.015,
            label = "99% VaR", hjust = 1, colour = "red", size = 3.5, fontface = "bold") +
  labs(title = "Bitcoin Daily Returns vs Historical VaR",
       x = "Date",
       y = "Daily Log Return") +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 16, hjust = 0.5),
    axis.text = element_text(size = 11),
    axis.title = element_text(size = 13)
  )

#Figure 2 - Ethereum GARCH-Predicted Volatility Overlay on Returns

# Remove NA and fit GARCH model
eth_returns <- na.omit(eth$return)

spec <- ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(1, 1)),
  mean.model = list(armaOrder = c(0, 0), include.mean = TRUE),
  distribution.model = "norm"
)

fit <- ugarchfit(spec = spec, data = eth_returns)

# Extract conditional volatility
volatility <- sigma(fit)

# Match it with dates
```

```r
eth_plot <- data.frame(
  Date = index(eth_returns),
  return = as.numeric(eth_returns),
  volatility = as.numeric(volatility)
)

ggplot(eth_plot, aes(x = Date)) +
  geom_line(aes(y = return), colour = "black", linewidth = 0.3) +
  geom_line(aes(y = volatility), colour = "purple", linewidth = 0.7) +
  labs(title = "Ethereum Daily Returns and GARCH(1,1) Volatility",
       y = "Daily Log Return and Conditional Volatility",
       x = "Date") +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 16, hjust = 0.5),
    axis.text = element_text(size = 11),
    axis.title = element_text(size = 13)
  )
```

Listing A.2: R script BTC Var Exceedance and ETH Volatility Prediction (using GARCH)

Python Code: LTSM Prediction

```python
import pandas as pd
import numpy as np
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense
from sklearn.preprocessing import StandardScaler

# Load your test data (btc_test_data.csv must be in the same folder)
```

```python
data = pd.read_csv('btc_test_data.csv')

# Separate features and target
X = data.drop(columns=['return']).values
y = data['return'].values

# Scale features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Reshape for LSTM [samples, timesteps, features]
X_scaled = X_scaled.reshape((X_scaled.shape[0], 1, X_scaled.shape
    [1]))

# Build LSTM model
model = Sequential()
model.add(LSTM(50, activation='relu', input_shape=(X_scaled.shape
    [1], X_scaled.shape[2])))
model.add(Dense(1))
model.compile(optimizer='adam', loss='mse')

# Train model
model.fit(X_scaled, y, epochs=50, batch_size=32, verbose=1)

# Predict
y_pred = model.predict(X_scaled).flatten()

# Save predictions to CSV
output = pd.DataFrame({'LSTM_Pred': y_pred, 'Actual': y})
output.to_csv('lstm_predictions.csv', index=False)
print("Done! Results saved to lstm_predictions.csv")
```

Listing A.3: Python script for LSTM Prediction (using BTC)

R Code: Model Comparison

```r
library(forecast)
library(quantmod)
library(rugarch)
library(randomForest)
library(caret)
library(xgboost)
library(reshape2)
library(ggplot2)

# Step 1: Load Cleaned Bitcoin Data
btc <- read.csv("BTC_Clean.csv")
btc$Date <- as.Date(btc$Date)

# Step 2: Sort by date and calculate log returns
btc <- btc[order(btc$Date), ]
btc$return <- c(NA, diff(log(btc$Close)))

# Step 3: Remove NA from first return
btc_returns <- na.omit(btc$return)


#ARIMA

# Step 1: Fit ARIMA model using auto.arima
arima_model <- auto.arima(btc_returns)

# Step 2: Generate in-sample predictions
arima_pred <- fitted(arima_model)

# Step 3: Compute MSE and RMSE
actual <- btc_returns
predicted <- as.numeric(arima_pred)
```

```r

# Align lengths (in case of mismatch)
n <- min(length(actual), length(predicted))
actual <- actual[1:n]
predicted <- predicted[1:n]

mse <- mean((actual - predicted)^2)
rmse <- sqrt(mse)

# Step 4: Output results
cat("ARIMA MSE:", round(mse, 5), "\n")
cat("ARIMA RMSE:", round(rmse, 5), "\n")

# Load and prepare data
btc <- read.csv("BTC_Clean.csv")
btc$Date <- as.Date(btc$Date)
btc <- btc[order(btc$Date), ]
btc$return <- c(NA, diff(log(btc$Close)))
btc_returns <- na.omit(btc$return)

# Train/test split (80/20)
split_idx <- floor(0.8 * length(btc_returns))
train <- btc_returns[1:split_idx]
test <- btc_returns[(split_idx + 1):length(btc_returns)]

# Compute MSE and RMSE
mse <- mean((test - predicted)^2)
rmse <- sqrt(mse)

cat("ARIMA Test MSE:", round(mse, 5), "\n")
cat("ARIMA Test RMSE:", round(rmse, 5), "\n")

#GARCH

```

```r
# Prepare train/test split for GARCH
train <- btc_returns[1:split_idx]
test <- btc_returns[(split_idx + 1):length(btc_returns)]

# GARCH(1,1) specification
spec <- ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(1,1)),
  mean.model = list(armaOrder = c(0,0), include.mean = TRUE),
  distribution.model = "norm"
)

# Fit on training data
fit <- ugarchfit(spec, data = train)

# Forecast for test period
fc <- ugarchforecast(fit, n.ahead = length(test))
predicted <- as.numeric(fc@forecast$seriesFor)

# Compute MSE and RMSE
mse <- mean((test - predicted)^2)
rmse <- sqrt(mse)

cat("GARCH Test MSE:", round(mse, 5), "\n")
cat("GARCH Test RMSE:", round(rmse, 5), "\n")

#LAG

# Create lag features (supervised learning)
btc_ml <- data.frame(
  return = btc_returns,
  lag1 = dplyr::lag(btc_returns, 1),
  lag2 = dplyr::lag(btc_returns, 2),
  lag3 = dplyr::lag(btc_returns, 3)
)
```

```r
btc_ml <- na.omit(btc_ml)

# Train/test split
split_idx <- floor(0.8 * nrow(btc_ml))
train <- btc_ml[1:split_idx, ]
test <- btc_ml[(split_idx + 1):nrow(btc_ml), ]

# Rolling ARIMA
train_series <- train$return
test_series <- test$return

arima_preds <- numeric(length(test_series))

for (i in 1:length(test_series)) {
  ts_train <- ts(as.numeric(train_series))
  model <- auto.arima(ts_train)
  fc <- forecast(model, h = 1)
  arima_preds[i] <- as.numeric(fc$mean)
  train_series <- c(train_series, test_series[i])
}

# ARIMA rolling forecast errors (out-of-sample, for Table 2)
mse_arima <- mean((test_series - arima_preds)^2)
rmse_arima <- sqrt(mse_arima)
cat("ARIMA Rolling Test MSE:", round(mse_arima, 5), "\n")
cat("ARIMA Rolling Test RMSE:", round(rmse_arima, 5), "\n")

# Random Forest model
set.seed(42)
rf_model <- randomForest(return ~ ., data = train)

# Predict
predicted <- predict(rf_model, newdata = test)

```

```r
# Evaluate
actual <- test$return
mse <- mean((actual - predicted)^2)
rmse <- sqrt(mse)

cat("Random Forest Test MSE:", round(mse, 5), "\n")
cat("Random Forest Test RMSE:", round(rmse, 5), "\n")

#XGBOOST

# Prepare data
train_matrix <- xgb.DMatrix(data = as.matrix(train[, -1]), label
    = train$return)
test_matrix  <- xgb.DMatrix(data = as.matrix(test[, -1]), label =
     test$return)

# Train XGBoost model
set.seed(42)
xgb_model <- xgboost(data = train_matrix, nrounds = 100,
    objective = "reg:squarederror", verbose = 0)

# Predict
predicted <- predict(xgb_model, newdata = test_matrix)

# Evaluate
actual <- test$return
mse <- mean((actual - predicted)^2)
rmse <- sqrt(mse)

cat("XGBoost Test MSE:", round(mse, 5), "\n")
cat("XGBoost Test RMSE:", round(rmse, 5), "\n")


#LSTM
```

```r

lstm_preds <- read.csv("lstm_predictions.csv")
mse <- mean((lstm_preds$Actual - lstm_preds$LSTM_Pred)^2)
rmse <- sqrt(mse)
cat("LSTM Test MSE:", round(mse, 5), "\n")
cat("LSTM Test RMSE:", round(rmse, 5), "\n")

# Create dataframe for plotting: actual + predictions
comparison_df <- data.frame(
  Date = btc$Date[(length(btc$Date) - length(test$return) + 1):
     length(btc$Date)],
  Actual = test$return,
  ARIMA = arima_preds,
  GARCH = as.numeric(ugarchforecast(fit, n.ahead = length(test$
     return))@forecast$seriesFor),
  RF = predict(rf_model, newdata = test),
  XGB = pmax(pmin(predict(xgb_model, newdata = test_matrix), 0.1)
     , -0.1),
  LSTM = lstm_preds$LSTM_Pred
)

# Reshape for plotting
long_df <- melt(comparison_df, id.vars = "Date", variable.name =
   "Model", value.name = "Return")

long_df$Type <- with(long_df, ifelse(Model == "Actual", "Actual",
                                     ifelse(Model %in% c("ARIMA",
                                         "GARCH"), "Traditional",
                                            ifelse(Model %in% c("
                                               RF", "XGB"), "
                                               Machine Learning",
                                                "Deep Learning"))
                                               ))

```

```r
# Plot all models vs actual
ggplot(long_df, aes(x = Date, y = Return, colour = Model, linetype = Model)) +
  geom_line(linewidth = 0.6) +
  labs(title = "Predicted vs Actual Bitcoin Returns by Model Group",
       y = "Return",
       x = "Date") +
  facet_wrap(~ Type, scales = "free_y", ncol = 1) +
  theme_minimal(base_size = 13) +
  scale_colour_manual(values = c(
    "Actual" = "black",
    "ARIMA" = "blue",
    "GARCH" = "purple",
    "RF" = "forestgreen",
    "XGB" = "orange",
    "LSTM" = "red"
  )) +
  scale_linetype_manual(values = c(
    "Actual" = "solid",
    "ARIMA" = "dotted",
    "GARCH" = "twodash",
    "RF" = "dotdash",
    "XGB" = "longdash",
    "LSTM" = "dashed"
  )) +
  theme(
    legend.title = element_blank(),
    strip.text = element_text(face = "bold", size = 13),
    plot.title = element_text(face = "bold", size = 16, hjust = 0.5)
  )

mse_table <- data.frame(
```

```
  Model = c("ARIMA", "GARCH", "Random Forest", "XGBoost", "LSTM"),
  MSE = c(0.00076, 0.00076, 0.00089, 0.00183, 0.00065),
  RMSE = c(0.2749, 0.2755, 0.2977, 0.0428, 0.02551)
)

cat("\nTable: MSE and RMSE for All Models\n")
print(mse_table)
```

Listing A.4: R script for Model Comparison (using BTC)

R Code: BTC against Sentiment

```
library(httr)
library(jsonlite)
library(ggplot2)

# Fetch historical Fear & Greed Index data
url <- "https://api.alternative.me/fng/?limit=0"
response <- GET(url)
raw_content <- content(response, as = "text", encoding = "UTF-8")
fng_data <- fromJSON(raw_content)$data

# Convert to data frame
fng_df <- data.frame(
  value = as.numeric(fng_data$value),
  classification = fng_data$value_classification,
  date = as.Date(as.POSIXct(as.numeric(fng_data$timestamp),
    origin = "1970-01-01"))
)

# Load Bitcoin data
btc <- read.csv("BTC_Clean.csv")
btc$Date <- as.Date(btc$Date)
```

```r
btc <- btc[order(btc$Date), ]
btc$return <- c(NA, diff(log(btc$Close)))
btc_returns <- na.omit(btc[, c("Date", "return")])

# Merge datasets on Date
merged_df <- merge(fng_df, btc_returns, by.x = "date", by.y = "Date", all = FALSE)

ggplot(merged_df, aes(x = date, y = value)) +
  geom_line(color = "steelblue") +
  labs(
    title = "Crypto Fear & Greed Index Over Time",
    x = "Date",
    y = "Index Value"
  ) +
  theme_minimal()

# Normalize data for comparison
merged_df$scaled_return <- scale(merged_df$return)
merged_df$scaled_fng <- scale(merged_df$value)

# Plot
ggplot(merged_df, aes(x = date)) +
  geom_line(aes(y = scaled_return, color = "Bitcoin Return")) +
  geom_line(aes(y = scaled_fng, color = "Fear & Greed Index")) +
  labs(
    title = "Bitcoin Returns vs. Fear & Greed Index",
    x = "Date",
    y = "Scaled Values",
    color = "Legend"
  ) +
  theme_minimal()
```

Listing A.5: R script for Comparing BTC vs Sentiment Index

R Code: RMSE Comparison

```r
library(ggplot2)
library(fmsb)

rmse_arima <- 0.02749 # from rolling ARIMA
rmse_garch <- 0.02755 # from original GARCH mean forecast
rmse_rf <- 0.02977 # from Random Forest
rmse_xgb <- 0.0428 # from XGBoost
rmse_lstm <- 0.02551 # from lstm_preds

model_performance <- data.frame(
  Type = c("Traditional", "Traditional", "Machine Learning", "Machine Learning", "Deep Learning"),
  Model = c("ARIMA", "GARCH", "Random Forest", "XGBoost", "LSTM"),
  RMSE = c(rmse_arima, rmse_garch, rmse_rf, rmse_xgb, rmse_lstm) # Replace with actual rmse vars
)

ggplot(model_performance, aes(x = Model, y = RMSE, fill = Type)) +
  geom_bar(stat = "identity", position = position_dodge()) +
  labs(title = "Figure 5: Model RMSE Comparison by Type",
       y = "Root Mean Squared Error",
       x = "Model") +
  theme_minimal(base_size = 13) +
  theme(legend.title = element_blank())
```

Listing A.6: R script for Comparison of RMSE of each model

R Code: Summary of Performance between Models

```r
library(httr)
library(jsonlite)
library(ggplot2)
library(zoo)
library(dplyr)
library(tidyr)

#run Model Comp before to get prediction comparison

# Fetch historical Fear & Greed Index data
url <- "https://api.alternative.me/fng/?limit=0"
response <- GET(url)
raw_content <- content(response, as = "text", encoding = "UTF-8")
fng_data <- fromJSON(raw_content)$data

# Convert to data frame
fng_df <- data.frame(
  value = as.numeric(fng_data$value),
  classification = fng_data$value_classification,
  date = as.Date(as.POSIXct(as.numeric(fng_data$timestamp),
     origin = "1970-01-01"))
)

# Load Bitcoin data
btc <- read.csv("BTC_Clean.csv")
btc$Date <- as.Date(btc$Date)
btc <- btc[order(btc$Date), ]
btc$return <- c(NA, diff(log(btc$Close)))
btc_returns <- na.omit(btc[, c("Date", "return")])

# Merge datasets on Date
merged_df <- merge(fng_df, btc_returns, by.x = "date", by.y = "
   Date", all = FALSE)

```

```r
# Ensure Date columns have same class
comparison_df$Date <- as.Date(comparison_df$Date)
merged_df$date <- as.Date(merged_df$date)

# Merge predictions by date
merged_df <- merge(
  merged_df,
  comparison_df[, c("Date", "ARIMA", "GARCH", "RF", "XGB", "LSTM")],
  by.x = "date",
  by.y = "Date",
  all.x = TRUE
)

# Calculate 21-day rolling volatility using the 'return' column
merged_df$Volatility <- zoo::rollapply(merged_df$return, width = 21, FUN = sd, align = "right", fill = NA)


# Assign volatility regimes
quantiles <- quantile(merged_df$Volatility, probs = c(1/3, 2/3), na.rm = TRUE)
merged_df$Vol_Condition <- with(merged_df,
                                ifelse(Volatility < quantiles[1], "Low",
                                       ifelse(Volatility > quantiles[2], "High", "Mid"))
)

# Assign sentiment regimes (from Fear & Greed index)
merged_df$Sentiment_Condition <- with(merged_df,
                                      ifelse(value <= 30, "Extreme Fear",
```

```
                                                ifelse(value >= 70,
                                                    "Extreme Greed",
                                                    "Neutral"))
)

# RMSE for each model and condition (Volatility)
long_vol <- merged_df %>%
  filter(!is.na(Vol_Condition)) %>%
  pivot_longer(cols = c(ARIMA, GARCH, RF, XGB, LSTM),
               names_to = "Model", values_to = "Prediction") %>%
  group_by(Vol_Condition, Model) %>%
  summarise(RMSE = sqrt(mean((Prediction - return)^2, na.rm =
     TRUE)), .groups = "drop")

# RMSE for each model and condition (Sentiment)
long_sent <- merged_df %>%
  filter(!is.na(Sentiment_Condition)) %>%
  pivot_longer(cols = c(ARIMA, GARCH, RF, XGB, LSTM),
               names_to = "Model", values_to = "Prediction") %>%
  group_by(Sentiment_Condition, Model) %>%
  summarise(RMSE = sqrt(mean((Prediction - return)^2, na.rm =
     TRUE)), .groups = "drop")

# Best model for each regime (Table 3 output)
best_vol <- long_vol %>% group_by(Vol_Condition) %>% slice_min(
    RMSE)
best_sent <- long_sent %>% group_by(Sentiment_Condition) %>%
    slice_min(RMSE)

# Print tables
cat("Table 3: Best Models by Volatility Regime\n")
print(best_vol)
cat("\nTable 3: Best Models by Sentiment Regime\n")
print(best_sent)
```

Listing A.7: R script for Model Performance Evaluation by Volatility and Sentiment